

GradeO: A Department-Aware Academic Performance Tracking and Student Information System

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Abstract - Academic performance data in many institutions is still handled through fragmented systems such as paper records, spreadsheets, institution-specific portals, or general learning management systems. These approaches often provide limited visibility, weak personalization, poor usability, and little support for continuous academic monitoring. GradeO is a full-stack academic performance tracking system designed to address these limitations through a centralized, student-oriented web application. The system provides secure login and signup, department-aware views, student records, test schedules, marks tracking, subject-wise total marks, password management, and academic information dashboards. The backend is implemented using Spring Boot with MySQL, while the frontend uses React with Vite for a modern responsive interface. GradeO emphasizes structured data access, role-aware information visibility, consistent academic categorization by PRN, and a cleaner user experience than traditional academic record systems.

Keywords - academic tracking, student information system, performance monitoring, React, Spring Boot, MySQL, educational software

I. Introduction

Student academic performance is influenced not only by classroom learning but also by how easily learners and institutions can access, interpret, and act on academic data. In many college environments, performance records are distributed across registers, spreadsheet files, notice-board-style portals, and isolated software tools. As a result, students often receive information only in fragmented form, while faculty and administrators face unnecessary effort in record maintenance and retrieval. GradeO was developed as a web-based academic tracking system that improves visibility, organization, and accessibility of student academic data. Rather than functioning only as a results portal, the system is designed as a usable academic workspace where students can view test records, marks, department-specific data, and consolidated performance information in one place.

II. Problem Statement

Existing academic record solutions commonly suffer from manual maintenance, inconsistent formatting, lack of centralization, poor progress visualization, weak personalization, and limited integration between marks, tests, and student records. These issues make it difficult for students to understand their academic progress and for institutions to manage information in a consistent and scalable way.

III. Objectives

The objectives of GradeO are to centralize academic information in a single platform, provide secure student access to marks and profile data, support department-wise and branch-wise filtering, present completed and upcoming tests clearly, summarize subject-wise performance using

scaled marks, and improve the academic user experience through a clean and responsive interface.

IV. Existing Solutions and Limitations

Traditional record-keeping methods such as physical registers are time-consuming, error-prone, and difficult to retrieve quickly. Spreadsheet-based approaches provide digital storage but remain isolated and highly dependent on manual interpretation. Learning management systems mainly focus on content delivery and assignment handling rather than end-to-end academic performance tracking. Institution-specific portals usually publish results with limited interaction and weak analytics, while simple mobile apps require manual input and lack institutional integration. GradeO improves on these approaches by combining centralized data access, secure identity-based visibility, department-specific filtering, and academic insights in a single system.

V. Proposed System

GradeO is a centralized academic performance tracking platform for students and academic stakeholders. The current system supports PRN-based signup and login, institutional email validation, secure session persistence, branch and department inference from student PRN, dashboard-based academic summaries, student list views filtered by department, test list views with status and date filtering, marks display for the logged-in student, subject-wise total marks analysis, password reset, and department-focused academic information views.

VI. System Architecture

GradeO follows a client-server architecture. The frontend is implemented using React and Vite, providing protected routes and interactive pages for dashboard, profile,

students, tests, marks, total marks, and department information. The backend is implemented using Spring Boot and exposes REST-style endpoints for authentication, marks retrieval, student data, test data, password reset, and schema overview. A MySQL relational database stores student, user, marks, tests, faculty, subject, and mapping data. This separation of concerns improves maintainability and enables modular development.

VII. Technology Stack

The frontend uses React 19, React Router, JavaScript, CSS, and Vite. The backend uses Java 17, Spring Boot, Spring Web MVC, Spring Data JPA, Spring Data JDBC, Spring Validation, and Spring Security Crypto. MySQL 8 is used as the database, while Maven and npm manage project dependencies and builds.

VIII. Functional Modules

The authentication module supports account creation, PRN-based login, secure password handling, and institutional email restriction. The academic classification module derives branch and department from PRN digits to keep user data consistent. The dashboard module presents test counts, faculty coverage, upcoming assessments, and progress indicators. The tests module supports department-specific filtering by subject, date, and status. The marks module displays student-specific marks securely, while the total marks module aggregates subject-wise total scored marks and total tested marks using scaled maximum marks. The information module filters subjects, faculties, and academic metadata by department. The profile module displays account details and supports password reset.

IX. Database Design Overview

The core database entities include student, user, test, subjects, faculty, faculty_subject, marks_scored, and component. These relations enable the system to connect students to assessments, associate tests with faculty and subjects, and retrieve marks with sufficient academic context. Seed data is used to maintain realistic branch-wise and department-wise datasets, including upcoming tests across all branches.

X. Security and Validation

The system validates PRN length and numeric format, enforces password length constraints, restricts signup to @sitpune.edu.in email addresses, uses BCrypt hashing for password storage, and limits marks access to the logged-in student except for authorized administrative views. Centralized exception handling ensures consistent API error reporting and improves robustness.

XI. Results and Discussion

The current implementation demonstrates that GradeO successfully centralizes students, tests, marks, and department-specific information into a single platform. It improves usability compared with spreadsheets and static portals by presenting structured dashboards, upcoming test

visibility, and subject-wise total marks summaries.

Department normalization across the frontend and backend also ensures that academic filters and displayed information remain consistent.

XII. Limitations and Future Scope

Although GradeO provides a solid academic tracking foundation, future enhancements can include attendance integration, semester-wise longitudinal analysis, analytics for performance prediction, role-specific faculty and administrator workflows, report generation, stronger audit logging, and cloud deployment for institution-scale usage.

XIII. Conclusion

GradeO demonstrates how a focused academic tracking system can improve the way student performance data is stored, accessed, and interpreted. By integrating a React frontend, Spring Boot backend, and MySQL database, the platform offers centralized academic visibility with secure access, department-aware filtering, and meaningful summaries of student performance. In its current form, GradeO represents a practical and extensible academic software solution suitable for project submission and further institutional development.

References

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